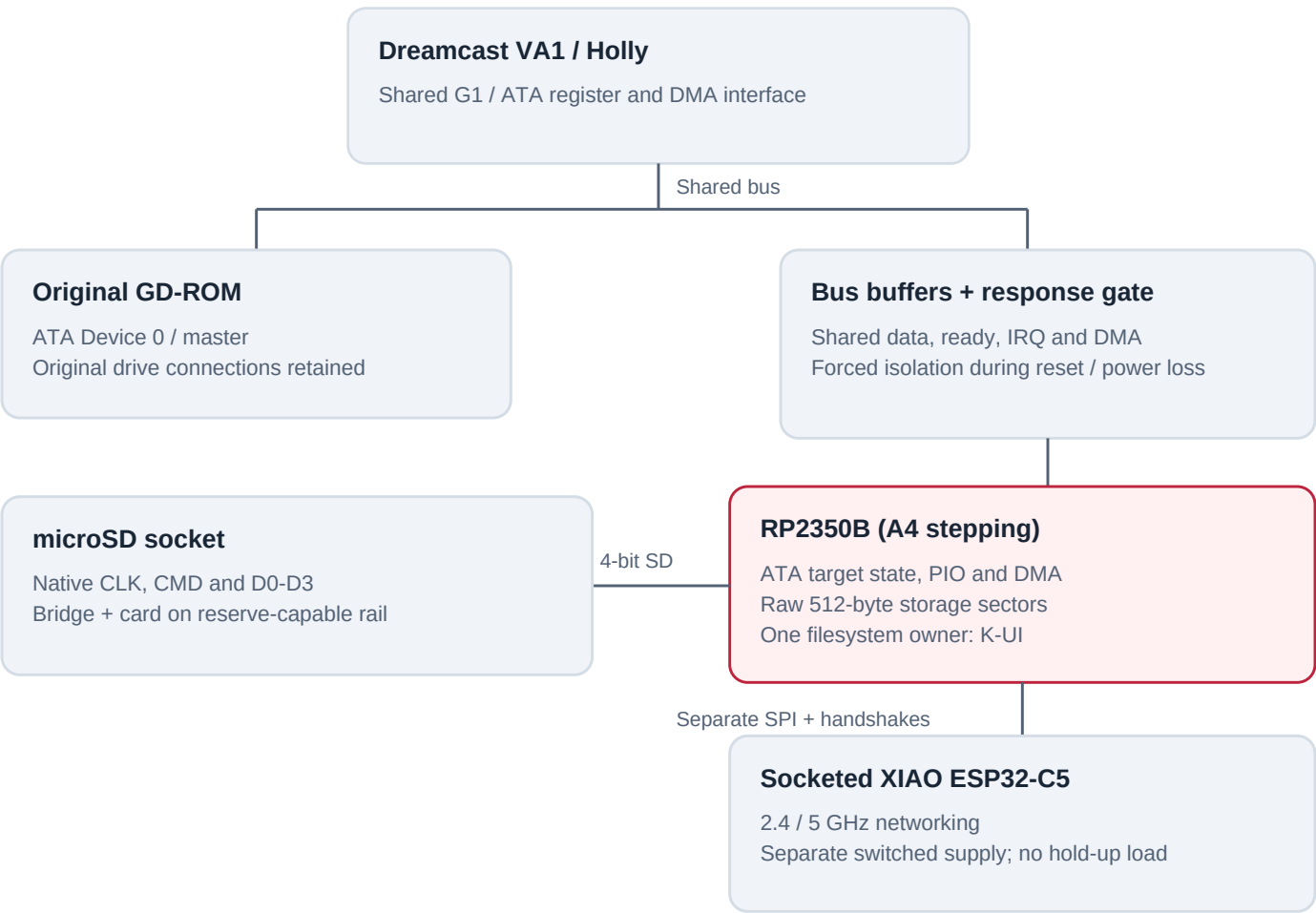


K-UI / G1 BOARD

01 Architecture

Proposed internal VA1 board. The original GD-ROM stays installed; the bridge is fixed as ATA Device 1 (slave).

Step 1 result: a complete candidate GPIO budget and a hardware block diagram. GPIO/peripheral placement has been checked; ATA timing, electrical gating and physical attachment remain open.



Design commitments

K-UI owns exFAT/ext4; the RP2350B exposes sectors and enforces write/flush policy. The C5 does not independently mount the game filesystem. Normal playback should write-protect the game partition after any required recovery.

Scope limits

No Dreamcast solder-point or connector numbering is approved here. No promised transfer rate, manufacturing files or stock-game network compatibility is implied. Additional glue logic may still be required after ATA timing review.

K-UI / G1 BOARD

02 G1 / ATA GPIO allocation

Numbers below are RP2350B GPIO identifiers, not QFN package pins, header positions or Dreamcast solder points. All assignments are proposed.

GPIO	Signal	RP direction	Purpose / constraint
0-15	ATA DD0-DD15	Bidirectional	Contiguous 16-bit data; through bus buffers
16	ATA_DA0	input	Register address bit 0
17	ATA_DA1	input	Register address bit 1
18	ATA_DA2	input	Register address bit 2
19	ATA_CS0n	input	Command register bank select
20	ATA_CS1n	input	Control register bank select
21	ATA_DIORn	input	Host read strobe
22	ATA_DIOWn	input	Host write strobe
23	ATA_RESETh	input	Reset sense; selectable path, behavior to qualify
24	ATA_IORDY	output / release	Ready / wait; only drive when permitted
25	ATA_INTRQ	output / release	Shared interrupt; selected-device rules apply
26	ATA_DMARQ	output / release	DMA request; selected-device rules apply
27	ATA_DMACKn	input	DMA acknowledge; required even with CS inactive
28	DATA_OEn	output	Data-buffer enable; default disabled
29	DATA_DIR	output	Data-buffer direction; exact polarity set by part
30	G1_RESP_EN	output	Proposed response gate; hardware interlock required
31	PFAILn	input	Early warning from unheld console supply

PIO plan: reserve PIO0 and PIO1 with GPIO base 0. The data bus and response controls remain in GPIO0-30. State-machine and instruction allocation still require a working ATA implementation.

Shared-bus rule: CS0/CS1 select register banks, not master versus slave. Track the DEV bit in the Device/Head register. DMA can use DMACK while both CS signals are inactive. G1_RESP_EN is an allocation placeholder, not a completed safe gating circuit.

K-UI / G1 BOARD

03 SD / C5 / diagnostic GPIOs

47 GPIOs assigned, one spare. The optional status LED can be removed if bus isolation needs another pin.

GPIO	Signal	RP direction	Connection / purpose
32	SD_CLK	output	Native four-bit SD clock; PIO2
33	SD_CMD	bidirectional	Native SD command; PIO2
34	SD_D0	bidirectional	SD data bit 0; PIO2
35	SD_D1	bidirectional	SD data bit 1; PIO2
36	SD_D2	bidirectional	SD data bit 2; PIO2
37	SD_D3	bidirectional	SD data bit 3; PIO2
38	SD_CDn	input	Socket card-detect switch, if provided
39	STATUS_LED	output	Optional; reclaimable if more isolation pins needed
40	C5_MISO	input	SPI1 RX -> XIAO D9 / C5 GPIO9
41	C5_CSn	output	SIO-controlled CS -> XIAO D0 / C5 GPIO1
42	C5_SCK	output	SPI1 SCK -> XIAO D8 / C5 GPIO8
43	C5_MOSI	output	SPI1 TX -> XIAO D10 / C5 GPIO10
44	C5_IRQn	input	Proposed packet IRQ -> XIAO D5 / C5 GPIO24
45	C5_READY	input	Proposed transaction-ready -> XIAO D4 / C5 GPIO23
46	C5_POWER_EN	output	Switched-supply enable; USB/back-power review needed
47	SPARE	unassigned	Reserve for extra isolation control or diagnostics

SD placement: PIO2 uses GPIO base 16 (window GPIO16-47). CLK32, CMD33 and consecutive data pins34-37 fit native four-bit SD PIO. Code that assumes GPIOs below32 must be adapted.

C5 placement: SPI1 is GPIO40 RX,42 SCK and43 TX; GPIO41 is manually held chip-select. D0, D4, D5 and D8-D10 are the proposed socket connections. IRQ and READY meanings are new firmware requirements, not existing K-UI C5 signals.

Power and programming: do not join the XIAO 3V3_OUT pin to the bridge regulator. Use a separately switched C5 supply with reviewed USB isolation. CHIP_EN is not on the standard socket, so power control replaces that reset connection. Keep dedicated QSPI flash, SWD, RUN and USB programming accessible; those are outside this GPIO table.

K-UI / G1 BOARD

04 What happens next

Complete the electrical decisions before turning this allocation into a schematic or ordering prototype boards.

Your next action

Install or open KiCad and create a project named **KUI-G1-Bridge-RevA**. The next hardware detail needed is the GD-ROM PCB manufacturer/model marking and the intended attachment method. Existing hole-reference photos are useful for mounting but do not establish connector orientation or drive-board identity.

Next design pass

Choose the VA1 G1 attachment and verify orientation/continuity; pick data/control buffers and selected-device gating; finish RP2350B power, flash, clock and programming circuits; add the native SD socket and socketed C5 power/interface circuit.

Review items before fabrication

1. Prove that the bridge releases every shared output when it is not entitled to respond.
2. Verify PIO register-response and DMA timing against actual G1 waveforms.
3. Resolve reset behavior; an existing G1 mod reports detection problems with its reset link.
4. Prevent USB/SPI and reserve power from back-feeding an unpowered module or console.
5. Size reserve power from measured card write/flush completion; no capacitor value selected yet.
6. Validate SD access, write protection and flush/recovery using a test card before a collection.

Primary references (clickable)

[RP2350 current datasheet: pin functions and silicon errata](#)

[Raspberry Pi SDK: PIO pin windows and peripheral functions](#)

[Seeed XIAO ESP32-C5: socket pins and power labels](#)

[Espressif C5: SPI slave timing and handshake guidance](#)

[KallistiOS: G1 ATA master/slave, sectors and DMA API](#)

[ATA draft: device selection, shared signals and DMA](#)

[DragonCity G1-ATA: implementation and reset-link caveat](#)

[Linux ext4: journal semantics and power-loss limitations](#)

[KiCad: Linux downloads](#)